Greta Dolcetti

gretadolcetti@gmail.com | gretadolcetti.github.io | linkedin.com/in/greta-dolcetti
github.com/gretadolcetti | Google Scholar

Technical Skills

Programming: Python, C++, Java, SQL, Bash
Frameworks & Tools: PyTorch, Hugging Face, scikit-learn, NumPy, Pandas, Docker, Kubernetes, Git, Linux, transformers, PEFT, vLLM, LangChain
Knowledge: Artificial Intelligence, Machine Learning, Prompt Engineering, Data Science
Research Areas: LLMs for Code Generation, AI Safety & Guardrails, Static Analysis, Abstract Interpretation
Languages: Italian (Native), English (Fluent)

Experience

Research Intern, AI Security & Privacy Jul. 2026 – Oct. 2026
IBM Research (Team lead: Mark Purcell — one of 2 interns selected from 200+ applicants) *Dublin, Ireland*
- Improved LLM generation of quantum programs by 87% on Qiskit Human Eval and Qiskit Quantum Katas by exploiting an iterative Instruct-Validate-Repair (IVR) loop and feedback from LintQ (first-author paper, NeurIPS 2026 SaTQuML workshop).
- Improved benchmarks, implemented a complete modular pipeline and integrated it in existing open-source tools.

Research Intern, AI Security & Privacy May 2025 – Sep. 2025
IBM Research (Team lead: Mark Purcell) — one of 2 interns selected from 200+ applicants *Dublin, Ireland*
- Shipped a production guardrail model detecting incorrect LLM function calls by training a LoRA adapter on Granite 3.3 8B; released by IBM in the public granite-3.3-8b-security-lib on Hugging Face.
- Hardened agentic systems against tool-misuse attacks by blue-teaming function-calling agents and designing the corresponding guardrails (first-author paper, AAAI 2026 TrustAgent workshop).
- Contributed to a synthetic-data pipeline for foundational agent guardrails, published at ICLR 2026.

Doctoral Researcher Sep. 2023 – Present
Ca' Foscari University of Venice *Venice, Italy*
- Raised the correctness and safety of LLM-generated code by 59% and 89% respectively by building a feedback loop that uses testing and static analysis as feedback for repairing generated code, published in *Discover Artificial Intelligence*.
- Built Pyra, a high-level linter that detects 16 data-science code smells that general-purpose Python linters miss; published in *Knowledge-Based Systems* and demoed at *ECML PKDD 2025*.

Visiting Researcher, ANTIQUE Group Aug. 2024 – Dec. 2024
Inria & École Normale Supérieure (Supervisor: Prof. Caterina Urban) *Paris, France*
- Cut the cost of verifying neural-network explanations up to an order of magnitude by co-developing a faster verification method for formally guaranteed explanations (ECOOP 2026).

Research Fellow Dec. 2021 – Mar. 2022
University of Parma *Parma, Italy*
- Predicted SARS-CoV-2 spike protein mutations by combining constraint programming with molecular modelling; the resulting workflow was published in *Molecules* (2023).

Selected Projects

Pyra *(Python, KBS 2026)* | Lead developer of a high-level linter that detects data-science code smells via abstract data types.
Granite 3.3 8B Function-Calling LoRA *(PyTorch, IBM 2025)* | Guardrail adapter flagging incorrect function calls from LLM agents; shipped in IBM's public model library.
Crab/Clam *(C++, SAS 2023 / STTT 2024)* | Contributor to a production abstract-interpretation framework: PPLite domain integration (PR #52, merged), split operator and unconstrained-variable oracles; 2.27x and 2.24x analysis speedups.

Invited Talks & Service

2025–2026 – Organizer of the Generative Code Intelligence Workshop at ECAI 2025 and IJCAI-ECAI 2026
2026 – Invited talk, National University of Singapore (NUS)
2025 – Invited talk, Dagstuhl Seminar 25421 on Sound Static Program Analysis in Modern Software Engineering
2023–26 – Artifact Evaluation Committee (VMCAI 2027, SAS 2026, VMCAI 2026, ECOOP 2024, ATVA 2024, SAS 2024, SAS 2023); Program Committee (AI4DT&CP @ IJCAI 2024)
2022–23 – Mentored high-school girls in STEM through the *Ragazze Digitali* coding program (96 hours)

Education

Ca' Foscari University of Venice Venice, Italy
Ph.D. in Computer Science *Sep. 2023 – Present*
- Thesis: *Static Analysis for Unconventional Targets - Analyzing LLM-Generated Code and Data Science Pipelines.* (supervisors: Professors Agostino Cortesi (Ca' Foscari), Enea Zaffanella and Vincenzo Arceri (University of Parma)). Implemented an autonomous generate-verify-repair pipeline that checks LLM-written code for correctness and vulnerabilities before deployment, and a static analyzer based on abstract interpretation that, through a domain of abstract types, annotates data science code and reports code smells caught neither at compile time nor at run time.

University of Parma Parma, Italy
M.Sc. in Computer Science — 110/110 cum laude *Sep. 2021 – Jul. 2023*

- Thesis: *Abstract Compilation Techniques for Static Analysis* (supervisor: Prof. Enea Zaffanella). Implemented two abstract compilation techniques in the Crab/Clam framework: the *split operator*, a new approach to filtering predicates over abstract domains that avoids the inefficiencies of the default procedure without losing precision; and *likely unconstrained variable oracles*, four implementations of an oracle that guesses which variables are unconstrained in CFG statements and guides the simplification of the abstract program, yielding significant speedups.

University of Parma Parma, Italy

B.Sc. in Computer Science — 110/110 *Sep. 2018 – Sep. 2021*

- Thesis: *Development of an Interface for Explainable AI for CAT Scan Analysis* (supervisor: Prof. Alessandro Dal Palù). Built a framework that explains in natural language why an Answer Set Programming reasoner retrieved a given set of arteries in a CAT scan, and lets users extend the knowledge base with new rules written in natural language and translated into AnsProlog.

Publications

17 publications, 8 as first author. Full list on Google Scholar.

NeurIPS 2026 (SaTQuML Workshop) | Linter-Guided Repair of Quantum Code Smells. https://openreview.net/forum?id=Q6uE2LmrH1

ICLR 2026 | Building a Foundational Guardrail for General Agentic Systems via Synthetic Data. arXiv:2510.09781

ECOOP 2026 | Faster Verified Explanations for Neural Networks. 10.4230/LIPIcs.ECOOP.2026.3

AAAI 2026 (TrustAgent Workshop) | Blue Teaming Function-Calling Agents. arXiv:2601.09292

NSAD 2026 | Quantum Computing and Static Analysis: State of the Art and Research Opportunities https://doi.org/10.1145/3840563.3843709

Knowledge-Based Systems 2026 | PYRA: A High-level Linter for Data Science Software. https://doi.org/10.1016/j.knosys.2026.115412

Discover Artificial Intelligence 2026 | Helping LLMs Improve Code Generation Using Feedback from Testing and Static Analysis. 10.1007/s44163-026-01009-5

ECML PKDD 2025 (Demo Track) | Introducing Pyra: A High-Level Linter for Data Science Software. 10.1007/978-3-032-06129-4_29

PeerJ Computer Science 2025 | Faster Numeric Static Analyses with Unconstrained Variable Oracles. https://doi.org/10.7717/peerj-cs.3390

Frontiers in Computer Science 2025 | A Dual Perspective Review on Large Language Models and Code Verification. https://doi.org/10.3389/fcomp.2025.1655469

NSAD 2024 | Towards a High Level Linter for Data Science. 10.1145/3689609.3689996

FTfJP 2024 | Towards a Sound Construction of EVM Bytecode Control-flow Graphs. 10.1145/3678721.3686227

STTT 2024 (SOAP Special Issue) | Speeding up Static Analysis with the Split Operator. 10.1007/s10009-024-00761-2

SAS 2023 | Unconstrained Variable Oracles for Faster Static Analyses. 10.1007/978-3-031-44245-2_5

SOAP @ PLDI 2023 | Speeding up Static Analysis with the Split Operator. 10.1145/3589250.3596141

Molecules 2023 | A Computational Workflow to Predict Biological Target Mutations: The Spike Glycoprotein Case Study. 10.3390/molecules28207082

ACS Medicinal Chemistry Letters 2022 | How a Blockchain Approach Can Improve Data Reliability in the COVID-19 Pandemic. 10.1021/acsmedchemlett.2c00077